

Scalability & Future Plan

Date	15 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application currently uses a single machine learning model (XGBoost) for loan prediction.	Prediction accuracy may be limited for diverse datasets.	High
2	User loan application details are not stored in a database.	Users cannot view previous loan prediction history.	Medium
3	The application is deployed on the Render free tier, which may experience startup delays after inactivity.	Users may notice slower response times during the first request.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of concurrent users.	Deploy on a scalable cloud platform with load balancing and auto-scaling support.
2	Data Storage	No database integration; predictions are processed without persistent storage.	Integrate MySQL or PostgreSQL to securely store user information and prediction history.
3	Performance	Suitable for small to medium workloads.	Optimize the Flask application, implement caching, and improve model inference efficiency.
4	Security	Basic input validation and secure deployment.	Add user authentication, role-based access control, HTTPS enforcement, and encrypted data storage.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Integrate a database to store user profiles and loan prediction history.	3–6 Months	Improved data management and user experience.
Phase 3	Add real-time credit score verification and banking API integration to enhance prediction reliability.	6–9 Months	More accurate loan eligibility assessment.
Phase 4	Develop a mobile application, multilingual support, analytics dashboard, and AI-powered loan recommendations.	9–12 Months	Increased accessibility, better user engagement, and enhanced decision support.